

Frontend - Single Person Role

ATDP | HTML/CSS/JS + Aston Martin Design System | 7 pages, Navi Mumbai demo

1. Overview & Scope

Full frontend ownership for ATDP (PROJECT_DOCUMENTATION.txt:4). 7 pages served statically via backend/server.js:31 at <http://localhost:5000> plus JSON API integration. Design: Aston Martin tokens (brand-primary #1B6157, racing-green palette, Cormorant Garamond 300 for display, Inter 400 for body, 0px/30px radii, 96px section gap) implemented in frontend/css/styles.css:4. No framework -- vanilla JS, Fetch API, localStorage.

2. Technology Stack

HTML5 semantic, CSS3 (custom properties, Flexbox, Grid, glassmorphism, preloader animations), JavaScript ES6+ (async/await, Fetch), Font Awesome 6, Chart.js (admin only), Inter + Cormorant Garamond via Google Fonts. Served via Express static, no build step.

3. Files Owned (frontend/*)

- frontend/index.html:1-222

Landing: sticky header 72px, hero 100vh photographic canvas + gradient overlay + IP banner, brand strip (local SVG logos /img/logos/*.svg via VehicleAPI.brands()), featured 6 vehicles (card-badge with brand logo), promo CTA band, 10 showrooms grid, 3-step How It Works (step-card hover to #1B6157), footer contact green #1B6157. Preloader phasing only on index (car icon + ATDP letters 0.6s, wings 1.2s, fade 1s, once/session via sessionStorage).

- frontend/vehicles.html:1-125

Catalog: filters Brand(Fetch)/Fuel/Max Price/Status (status default available, All), Search removed (dropdowns only per request), category=car forced, fuel_type server-side (was client-side bug), cleanParams for All Brands (fixes brand_id=undefined -> 0). Grid 3 cards with brand logo badge, pagination 9/page.

- frontend/vehicle.html:1-179

Detail + Booking Wizard 3 steps (Date -> Time Slot -> Confirm): auto-showroom box, date min tomorrow, slot-grid 9 slots 09-18, enforceLogin() banner + redirect login.html?redirect=... with User/Admin tabs, admin_message in confirm. Handles vehicle.status available check.

- frontend/login.html:1-90

Login with User/Admin tabs, Admin security 1234 (frontend check, hint removed per request), role validation vs JWT role, redirect support, demo creds John/Admin without revealing 1234.

- frontend/register.html:1-53

Register: name,email,password (8 chars + upper/lower/digit via backend validator), phone, address -> POST /api/auth/register -> JWT to dashboard.

- frontend/dashboard.html:1-134

Customer dashboard: green gradient section, bold stats (Total/Pending/Approved/Completed) on #1B6157, bookings table Status & Admin Response (green/red left border), admin_message congrats/reason, cancel + view, profile + change-password forms.

- frontend/admin.html:1-301

Admin dashboard: green stats, doughnut + bar Chart.js, tabs Bookings/Vehicles/Showrooms/Users, bookings table with vehicle_price (fix v.price), modal window (overlay 0.48, white card 560px, hairline-soft border, shadow, eyebrow brand-primary, textarea, Approve/Reject via BookingAPI.update with admin_message). Vehicle/showroom/user CRUD forms.

- frontend/css/styles.css:1-306

Tokens: brand-primary #1B6157, bright #00B497, surfaces, hairlines, text-ink/muted, radii none/xs/pill, spacing xxs-section, typography display-xl 54px w300 etc, preloader, header, hero, cards, filters, wizard, table, stats, footer green, step-card hover, toast, responsive 320/768/1025/1920 breakpoints.

- frontend/js/api.js:1-47

API wrapper: api(path,opts) with Bearer token, cleanParams() filtering undefined/null/empty before URLSearchParams (fixes All Brands 0 bug), AuthAPI, VehicleAPI.list/brands/get, ShowroomAPI, BookingAPI.list/slots/create/update, AdminAPI.dashboard/users/bookings, formatINR en-IN.

- frontend/js/app.js:1-79

Global: toast(), initPreloader() only on index.html + sessionStorage once, initHeader() hamburger, auth state (atdp_token/user in localStorage), userNav/mobileAuth, requireAuth(), IP banner via api.ipify.org.

- frontend/img/logos/*.svg:1

10 local SVG logos (maruti,hyundai,tata,mahindra,honda,toyota,kia,mg,bmw,mercedes) 120x40 brand colors, avoids external CDN 400.

4. Key User Flows Implemented

Customer: Visit -> Browse (filters) -> Register/Login (forced before booking) -> Select vehicle (showroom auto) -> Pick date -> Available slots (real-time) -> Confirm -> Dashboard (status pending/approved/cancelled + admin_message)

Admin: Login as Admin (1234) -> Dashboard analytics -> Manage vehicles/showrooms (add/delete) -> Approve/Reject bookings via modal (congrats vs reason) -> View users.

5. Design System (Aston Martin)

Colors: brand-primary #1B6157 CTA, bright #00B497 preloader, dark-teal #005A4C, accent-gold #B58C67, surface-canvas #161A11, white page, hairline #C4C4C4. Typography: Flare (Cormorant) w300 for display (72/54/48/36/26), Sans (Inter) 14px w400 body, 11px uppercase tracking 1.5px. Layout: 1200px max, 96px section gap, 0px cards, pill CTA 30px. Hover: step-card -> green solid, white text.

6. Effort & Timeline (solo frontend)

Structure + CSS tokens 2d, Landing + hero/preloader 2d, Catalog (filters/pagination) 2d, Detail/wizard 2d, Auth pages 1d, Dashboards (customer/admin + Chart.js + modal) 3d, Integration + responsive + rate-limit fix 2d => ~12-14 days. Parallel backend saves time, total 3-4 weeks solo.

7. Skills Required

HTML5, CSS3 (custom properties, Grid/Flex, animations), vanilla JS (Fetch, async/await, DOM), REST integration, localStorage, responsive design, basic Chart.js. Design eye for Aston Martin restraint (w300, photographic canvases, gradient overlays).

8. Run & Test

cd backend && npm install && node seed.js && node server.js # http://localhost:5000

Browse: / (hero), /vehicles.html (Brand/Fuel/Price/Status filters -> 28 cars, Hybrid 2, Petrol 4), /vehicle.html?id=1 (wizard), /login.html (User/Admin tabs), /dashboard.html, /admin.html (modal)

Test: Hard-refresh Cmd+Shift+R after CSS/JS changes (cleanParams, logos).